



DIGITAL COMPUTING ARCHITECTURES

MP | MCU | GPU | ASIC | FPGA

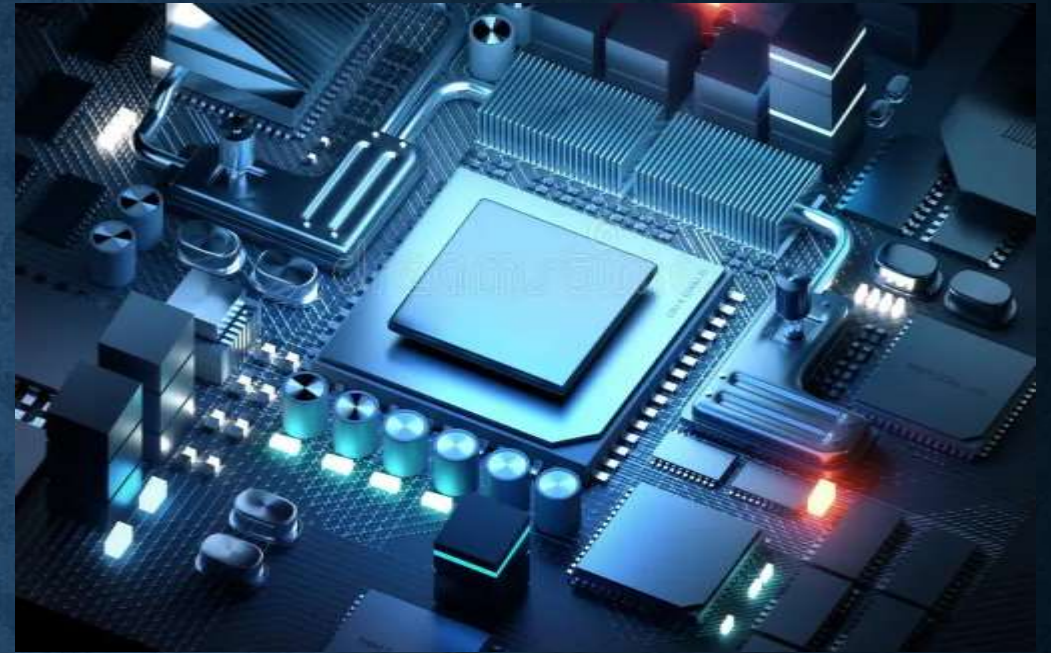
WHAT WE'LL COVER??

Explore four fundamental digital computing architectures that power Modern technology. From general-purpose processors to custom silicon , we'll dive deep Into:

1. Microprocessors
2. Microcontrollers
3. ASIC
4. FPGAs
5. Understanding their design , strengths ,and real-world application



MICRO- PROCESSOR



MICRO-PROCESSOR

A microprocessor is a Central Processing Unit(CPU) integrated onto a single semiconductor chip. It performs arithmetic ,logic ,and control Operations by executing stored program instructions. It requires external components to function as a complete system, including:-

1. RAM for volatile memory.
 2. ROM for firmware storage.
 3. I/O interfaces for communication with peripherals.
- Unlike a microcontroller ,It does not have memory and peripherals integrated on the same chip. It is designed for high computational performance and is commonly used in computers and Workstations. Examples include the Intel Core and AMD Ryzen processor families.



ADVANTAGES AND DISADVANTAGES OF MICROPROCESSORS.

ADVANTAGES:-

- Very high processing speed
- Can run operating systems
- Suitable for multitasking
Handles complex
computations
- Large memory support

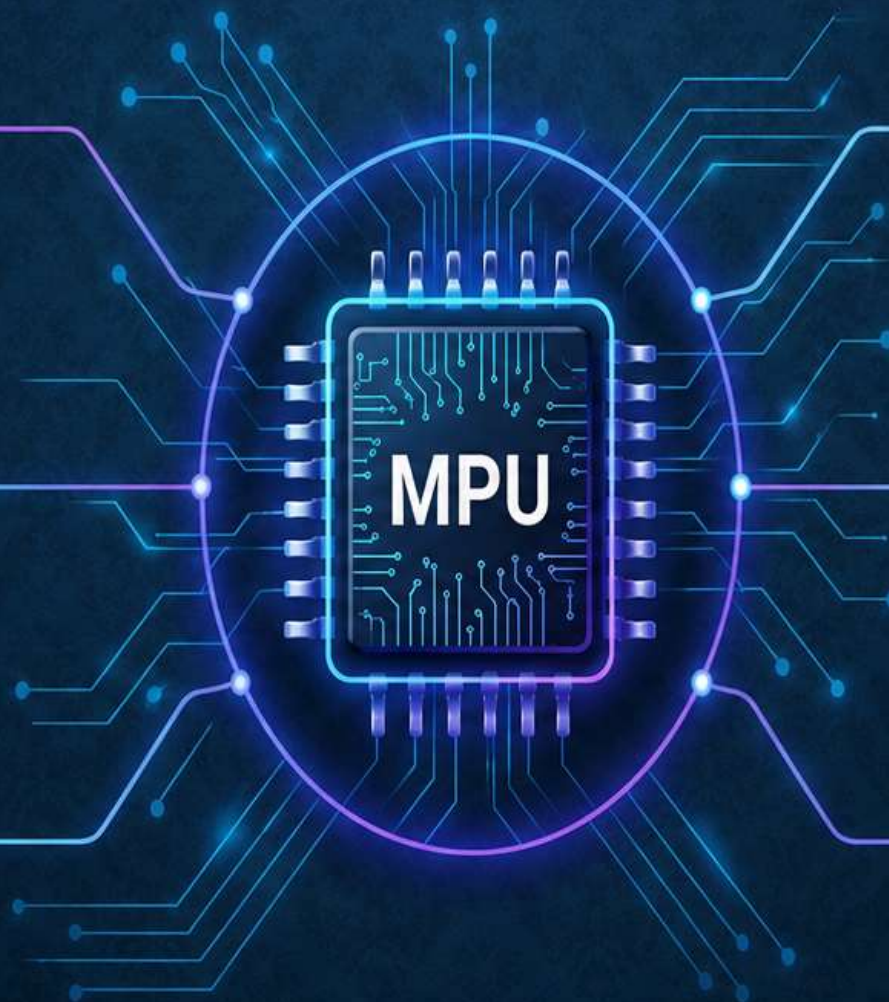
DISADVANTAGES:-

- Requires many external
components
- Higher power consumption
Larger PCB size
- Higher overall system cost

APPLICATIONS OF



MICROPROCESSOR



PERSONAL COMPUTERS (PCs)

Powers desktop, laptop, and all-in-one PCs for general computing, software execution, and internet access.



WORKSTATIONS & SERVERS

Used in high-performance workstations for specialized applications (e.g., video editing) and powerful data center servers.



SCIENTIFIC COMPUTING

Enables massive-scale simulations, weather forecasting, and genetic research in supercomputers.



GAME CONSOLES

Drives realistic graphics, complex AI, and high-performance physics in modern gaming systems.



SMARTPHONES & TABLETS

The main CPU of mobile devices, handling everything from apps and web browsing to cameras and operating systems.

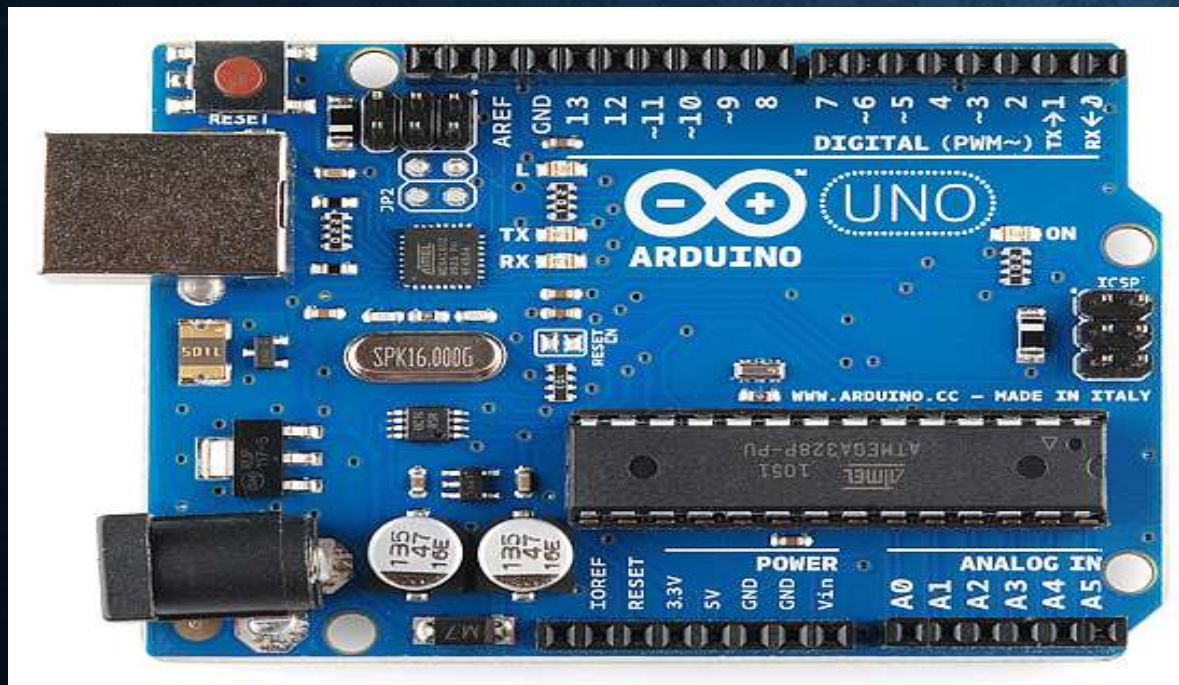
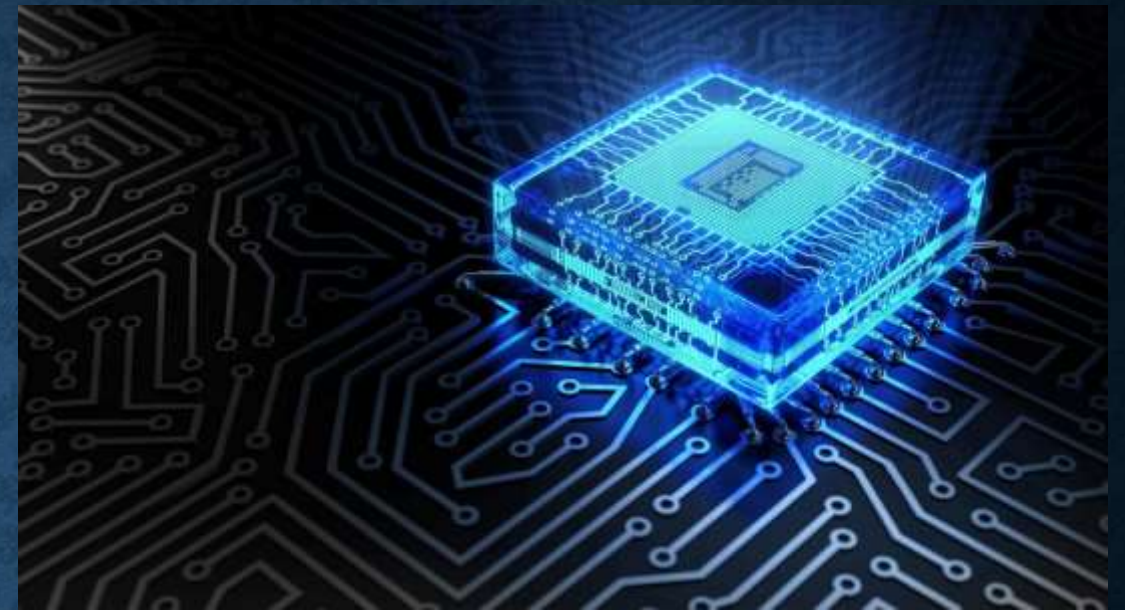


NETWORKING & CLOUD INFRASTRUCTURE

Crucial for high-speed routers, complex switches, and cloud-based data processing and services.

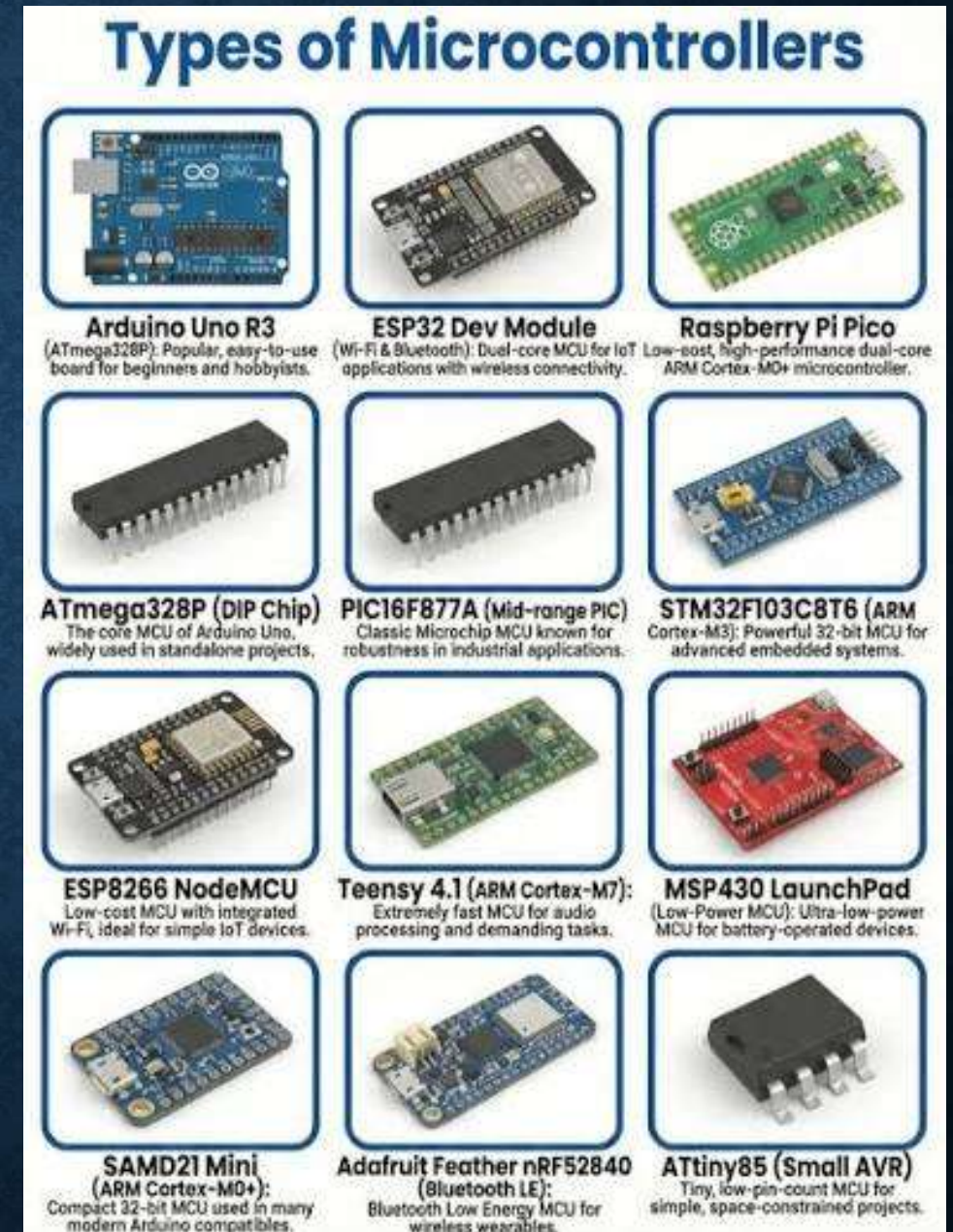


MICRO- CONTROLLER

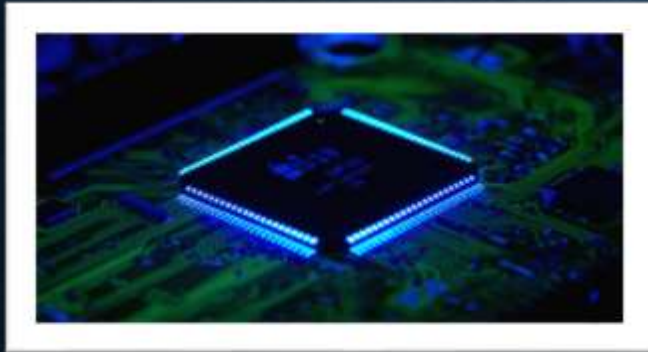


WHAT IS A MICRO-CONTROLLER?

- microcontroller (MCU) is a compact integrated circuit that combines a CPU , memory (RAM & Flash), and programmable input/output peripherals all on a single chip . Designed for embedded control applications , MCUs are self-contained computing systems ideal for dedicated tasks.
- Popular examples include Arduino (AVR/ARM), ESP32 (dual-core Xtensa), and STM32 (ARM Cortex-M)- powering billions of smart devices worldwide.



KEY FEATURES OF A MICROCONTROLLER



1.CPU + Memory +
Peripherals



2.Cost-effective and easy
to interface it with sensors



3. Low power consumption

ADVANTAGES AND DISADVANTAGES OF MICRO-CONTROLLER

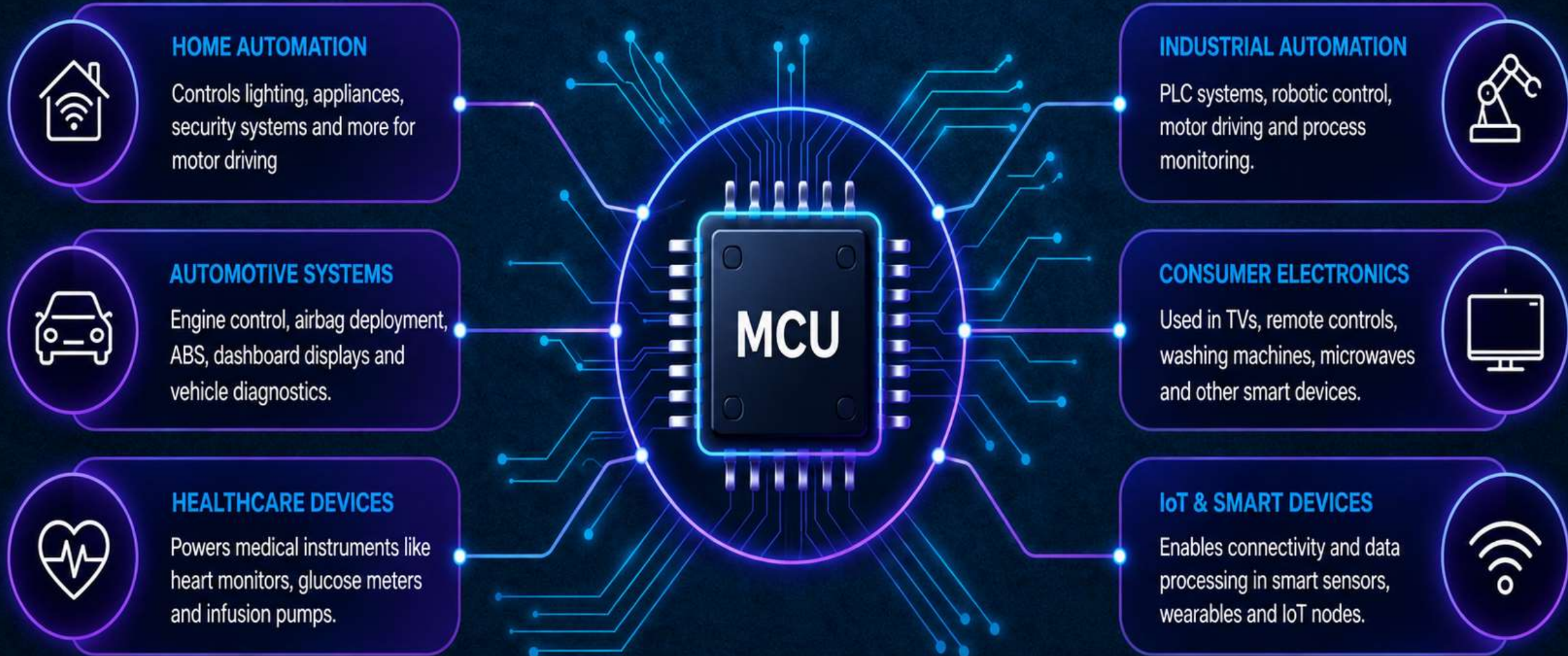
ADVANTAGES:-

- Low cost.
- Low power consumption Compact design.
- Easy to program.
- Fewer external components

DISADVANTAGES:-

- . lower processing power.
- Limited RAM.
- Limited Flash memory.
- Not suitable for heavy computation

APPLICATIONS OF MICROCONTROLLER



GPU (GRAPHIC PROCESSING UNIT).

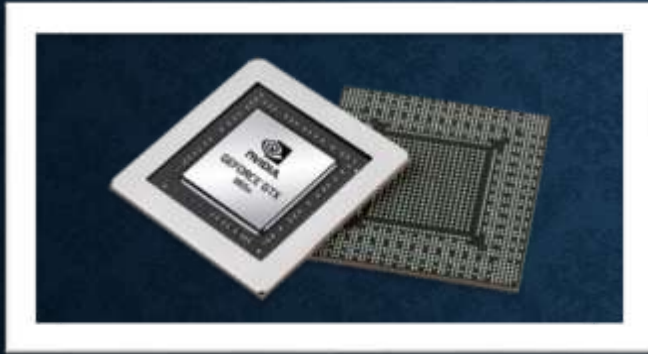
- Definition of GPU:-

A **GPU** is a massively parallel processor optimized for high-throughput computing. It employs a **Single Instruction, Multiple Data (SIMD)** or **Single Instruction, Multiple Threads (SIMT)** execution model, allowing thousands of lightweight threads to execute the same instruction on different data simultaneously. GPUs are designed to maximize computational throughput and memory bandwidth for parallelizable tasks.

- EXAMPLES:-
 1. NVIDIA GeForce RTX 4090
 2. AMD Radeon RX 7900 XTX
 3. Intel Arc Graphics B850



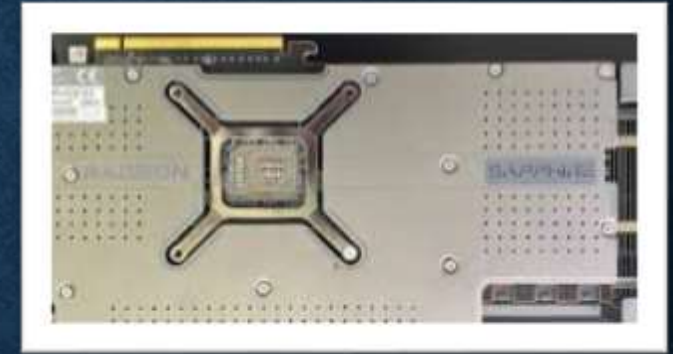
KEY FEATURES OF GPU



1. Thousands of Tiny
Cores



2. (High Bandwidth)



3. Specialized Matrix
and Hardware
Accelerators

ADVANTAGES AND DISADVANTAGES OF GPU.

ADVANTAGES:-

- 1.Massive Parallel Processing
- 2. High Computational Throughput
- 3. Faster Graphics Rendering
- 4. Excellent for AI and Machine Learning

DISADVANTAGES:-

1. Higher Power Consumption
2. Expensive
3. Complex Programming
4. Not Suitable for General System Control

APPLICATIONS OF GRAPHICS PROCESSING UNIT

Powering Parallel Performance for a Visual and Intelligent World



GAMING & 3D GRAPHICS

Renders high-quality visuals, real-time effects and immersive gaming experiences.



AI & MACHINE LEARNING

Accelerates training and inference for deep learning, neural networks and data analytics.



SCIENTIFIC & ENGINEERING SIMULATIONS

Speeds up weather forecasting, molecular modeling, physics and space research.

GPU

VIDEO & CONTENT CREATION

Boosts video editing, 4K/8K rendering, animation and live streaming with fast parallel processing.



DATA CENTERS & CLOUD

Powers HPC, virtual desktops, cloud gaming and AI-as-a-Service platforms.



AUTONOMOUS & SMART SYSTEMS

Enables self-driving cars, robotics, drones and real-time computer vision applications.

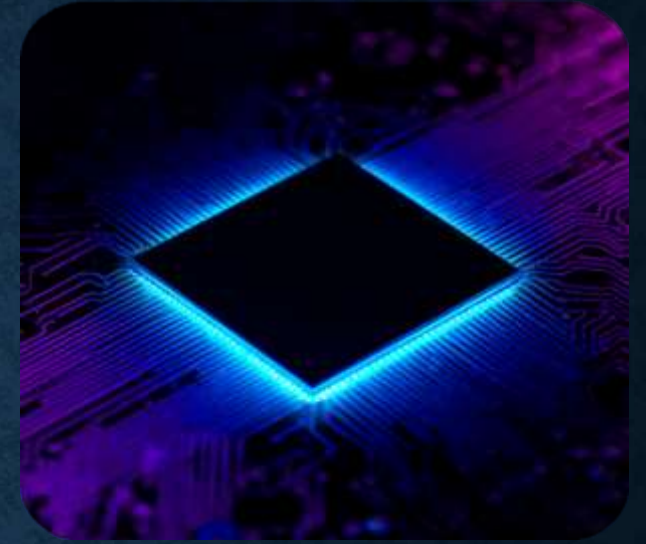


**ASIC(APPLICATION
SPECIFIC INTEGRATED
CIRCUIT).**

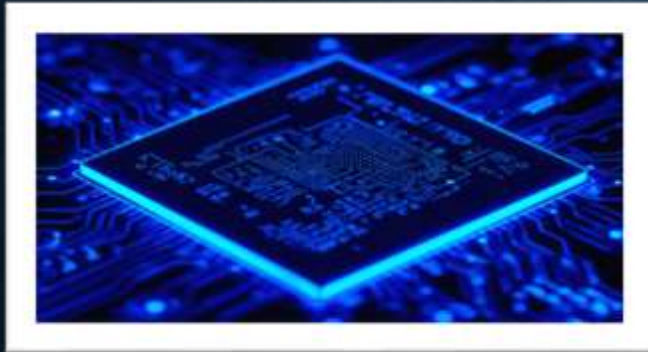


WHAT IS ASIC?

- Application-Specific Integrated Circuit (ASIC) is a Microchip custom-designed and manufactured to perform one specific function. Unlike general-purpose processors, Every transistor on an ASIC is optimized for its dedicated task — delivering unmatched speed, power efficiency, and performance. ASICs power everything from Bitcoin mining rigs to smartphone modems and AI accelerators.

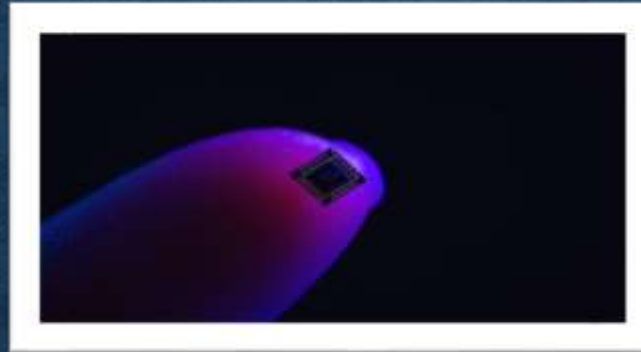


KEY FEATURES OF ASIC



1.Fixed hardware

2.High speed



3.Low power

4. High reliability



5.Optimised for one
single application

ADVANTAGES AND DISADVANTAGES OF ASIC

ADVANTAGES:-

1. Highest performance
2. Lowest power consumption
3. Small chip size
4. High reliability
5. Best for mass production

DISADVANTAGES:-

1. Very expensive development
2. Long design cycle
3. Cannot be reprogrammed
4. Errors require new fabrication

APPLICATIONS OF ASIC



1. CONSUMER ELECTRONICS

Used in smartphones, smartwatches, cameras and gaming devices for high performance and low power.



2. COMMUNICATION SYSTEMS

Power base stations, routers, 5G/6G infrastructure and network switches for fast data processing.



3. AUTOMOTIVE APPLICATIONS

Enable ADAS, engine control units, infotainment systems and electric vehicle management.



4. MEDICAL DEVICES

Used in MRI, pacemakers, hearing aids and diagnostic equipment for reliable precision and efficiency.



5. AI & MACHINE LEARNING

Accelerate deep learning inference in edge devices, data centers and AI accelerators.



6. DATA CENTERS & CLOUD

Improve performance and energy efficiency in servers, storage systems and cloud computing workloads.



7. SECURITY & ENCRYPTION

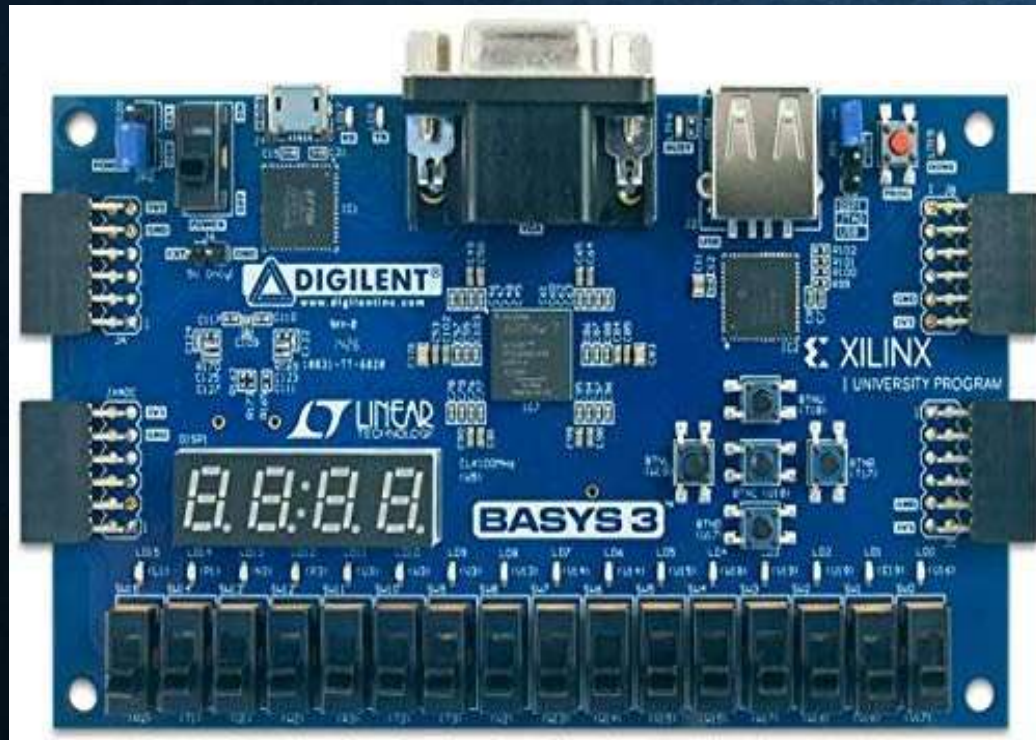
Provide hardware-level security, encryption, secure boot and trusted execution environments.



8. INDUSTRIAL & DEFENSE

Used in robotics, industrial automation, aerospace, and defense systems for high reliability and rugged performance.

FPGA(FIELD PROGRAMMABLE GATE ARRAY).



WHY IS IT CALLED FPGA?

Each letter in FPGA represents a core concept behind this powerful and flexible technology.

FPGA stands for Field-Programmable Gate Array — a semiconductor device whose logic is defined by the user post-manufacture. Unlike fixed ASICs, FPGAs can be reprogrammed repeatedly using HDLs such as Verilog or VHDL, making them ideal for prototyping and custom hardware design.



F — Field

Programmed by the user in the field — after manufacturing, not at the factory.



P — Programmable

Fully configurable and reconfigurable to implement any digital logic design.



G — Gate

Contains millions of logic gates enabling complex hardware computation.



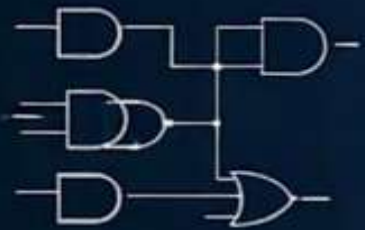
A — Array

Logic blocks arranged in an organized, interconnected array structure.

EVOLUTION OF FPGA.

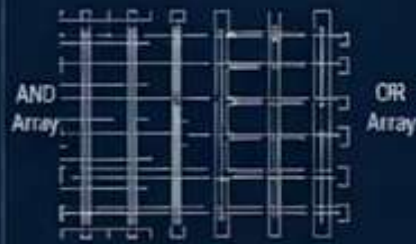
FROM EARLY DEVICES TO MODERN INTELLIGENT PLATFORMS

1. 1960s–70s DISCRETE LOGIC ERA



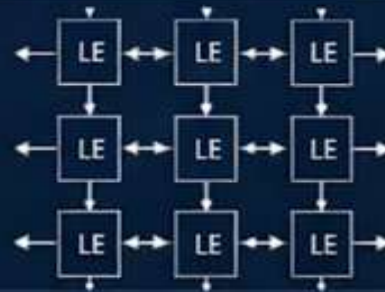
- Built with AND, OR gates
- Large, bulky, and power-hungry
- Designed for specific functions
- Long development time

2. 1980s PLD ERA



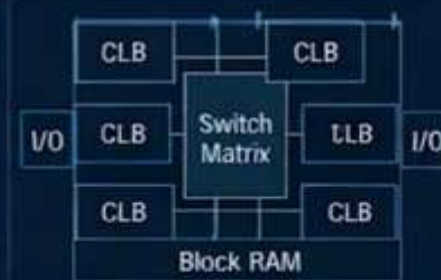
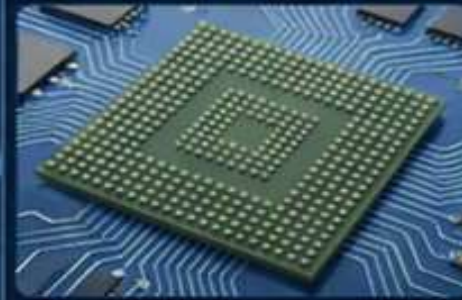
- Introduction of PLDs
- PROM, PAL, PLA devices
- Smaller, programmable
- Limited capacity and flexibility

3. 1990s CPLD ERA



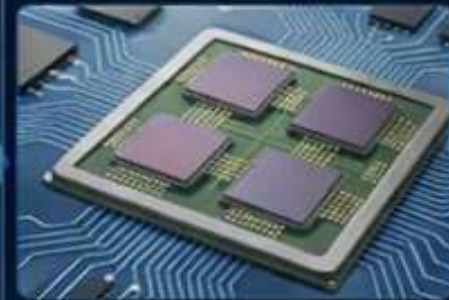
- CPLDs introduced
- High density, non-volatile
- Predictable timing
- Used for glue logic, control

4. Early 2000s FPGA ERA



- Rise of SRAM-based FPGAs
- Complex logic, DSP blocks, Block RAM
- Reconfigurable and flexible
- Used in communication, defense, and compute

5. 2010s–Today MODERN INTELLIGENT FPGA



- Millions of logic cells
- ARM cores, AI engines, DSPs, High-speed transceivers
- Hardware-software co-design
- Used in AI, 5G, automotive, cloud, and edge

Growth of FPGA

Feature evolution by decade — from embedded RAM to AI accelerators and beyond.



1990s: Early Enhancements

Embedded RAM blocks, hardware multipliers, and clock management units introduced, enabling more complex and efficient digital designs.



2000s: Advanced Integration

Embedded processors, dedicated DSP slices, and high-speed serial transceivers brought system-on-chip capabilities to FPGA platforms.



Today: Millions of Gates & AI

Modern FPGAs feature millions of logic gates, integrated ARM processors, AI/ML accelerators, and advanced security and networking features.

KEY FEATURES OF FPGA.



1.Reprogrammable

2. Parallel processing



3. Custom hardware

4.Fast prototyping



5.Real time operation

APPLICATIONS OF FPGA



1. COMMUNICATION SYSTEMS

Used in 5G/6G, wireless, satellite communication, routers and network acceleration.



2. VIDEO & IMAGE PROCESSING

Real-time video processing, image filtering, compression, and high-speed display systems.



3. AUTOMOTIVE APPLICATIONS

ADAS, sensor fusion, LIDAR, infotainment systems, and electric vehicle control.



4. INDUSTRIAL AUTOMATION

Motor control, robotics, PLC acceleration, machine vision, and factory automation.



5. MEDICAL & HEALTHCARE

Medical imaging, patient monitoring, signal processing and diagnostic equipment acceleration.



6. AEROSPACE & DEFENSE

Radar systems, avionics, secure communication, and mission-critical applications.



7. FINANCIAL TECHNOLOGY

High-frequency trading, risk analysis, real-time data processing and algorithmic acceleration.



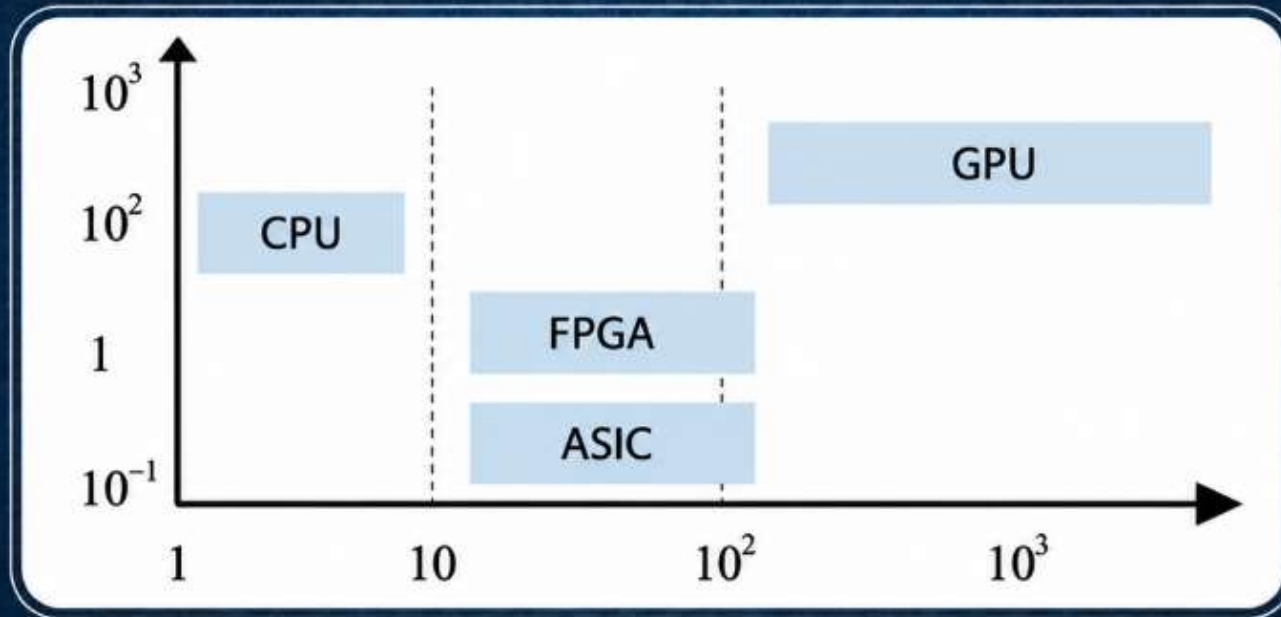
8. DATA CENTERS & HPC

Server acceleration, data compression, encryption and high-performance computing.

COMPARISON BETWEEN ARCHITECTURES



THROUGHPUT AND POWER COMPARISON



X-axis:- throughput (the amount of data the hardware system can successfully process within specific timeframe).

Y-axis:-power (electrical energy required by a hardware system to process data, measured in Watts (W)).

Metric / Spec	Microcontroller (MCU)	Microprocessor (CPU)	GPU	FPGA	ASIC
Typical Power (Watts)	<0.5W	15W - 350W	150W - 700W+	5W - 60W	<1W (edge) to 400W (cloud)

Which Architecture Should You Choose?

A quick decision path from requirement to recommendation

Need a full OS &
general-purpose computing?

YES



Choose MP

Cost & power sensitive,
simple embedded task?

YES



Choose MCU

Mass production
(100k+ units), fixed function?

YES



Choose ASIC

Need flexibility,
prototyping or reprogrammability?

YES



Choose FPGA

Need massively parallel
processing (AI, graphics,
simulation, data processing)?

YES



Choose GPU

ARCHITECTURE COMPARISON TABLE(SUMMARY).

SIDE-BY-SIDE COMPARISON

How the five architectures stack up across key parameters

PARAMETER	 MP	 MCU	 ASIC	 FPGA	 GPU
 Core Components	CPU only	CPU + RAM + ROM + I/O	Fixed custom logic	Programmable logic blocks	Massively parallel processing cores
 Flexibility	High (software)	Medium (software)	None (fixed function)	High (hardware reprogrammable)	High (software)
 Processing Speed	Very High	Low – Medium	Highest (for its task)	High	Very High (parallel)
 Power Consumption	High	Low	Lowest	Medium – High	High
 Development Cost	N/A (off the shelf)	Low	Very High	Medium	High
 Cost per Unit at Volume	High	Low	Lowest at scale	Higher than ASIC	High
 Reprogrammable?	Software only	Software only	No	Yes (hardware)	Yes (software)
 Time to Market	Immediate	Fast	Slow (months–years)	Fast	Fast
 Typical Use Case	PCs & servers	Embedded / IoT	Mass-produced chips	Prototyping & custom logic	AI, ML, gaming, HPC, parallel computing
 Best Suited For	General-purpose computing	Control, low-power embedded tasks	High-volume, specific applications	Custom logic, rapid prototyping	Data-parallel workloads, AI & scientific computing
 GPU / Parallel Capability	Limited (few cores)	None – Very Limited	Task-specific parallelism (optimized)	Moderate (custom parallel pipelines)	Excellent (thousands of cores, SIMD architecture)